Global Electronics Retailer Performance Analysis | Revenue, Customers & Market Trends

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1. Project Objective

The objective of this project is to perform **end-to-end sales data analysis** using:

- Microsoft Excel / Power Query → Data Cleaning & Transformation
- Power BI → Data Modelling, Visualization & Business Insights

The analysis focuses on:

- Revenue performance trends
- Customer behaviour patterns
- Product category performance
- Geographic sales distribution
- Business growth and risk indicators

The goal is to convert raw transactional data into **actionable business insights for decision-making**.

2. Dataset Overview

The dataset consists of four primary tables:

Fact Table

- **Sales**
    - Order Number
    - Order Date
    - Customer Key
    - Product Key
    - Quantity

Dimension Tables

- **Customers Data**
  - Gender, Country, City, State, Continent
- **Products**
  - Product Name, Category, Subcategory, Price
- **Stores**
  - Country, State, Store Size, Open Date

3. Problem Statement

Businesses often struggle with:

- Identifying **high-performing products and categories**
- Understanding **customer purchasing behaviour**
- Tracking **revenue trends over time**
- Detecting **regional performance gaps**
- Evaluating **sales efficiency vs quantity sold**

This project aims to:

- Analyze sales trends and performance
- Identify key revenue drivers
- Understand customer and geographic distribution
- Provide insights for strategic business decisions

4. Data Preprocessing

Data Cleaning

- Removed duplicate records
- Handled missing values
- Standardized formats (dates, categories)

Transformation

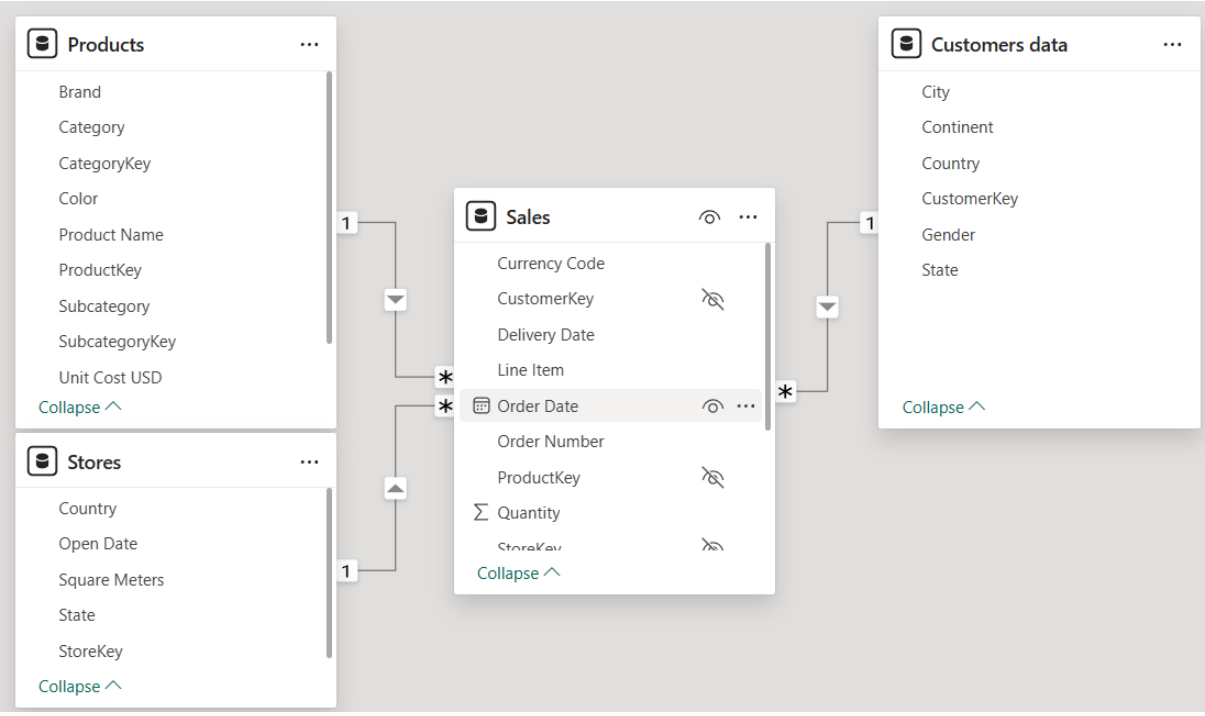
- Created calculated measures:
    - Total Revenue
    - Total Orders
    - Total Quantity
    - Average Order Value

Filtering & Structuring

- Organized into fact & dimension tables
- Prepared for star schema modelling

5. Data Modelling (Power BI)

A STAR SCHEMA MODEL was implemented:



- **Sales (Fact Table)** → Connected to:
    - Customers
    - Products

    - Stores

Relationships

- One-to-Many relationships
- Keys used:
    - Customer Key
    - Product Key
    - Store Key

👉 This improves:

- Query performance
- Drill-down capability
- Dashboard responsiveness

6. DAX Measures & KPI Development

After building the data model, key performance indicators (KPIs) and measures were created using DAX (Data Analysis Expressions) to support business analysis.

Key Measures Created

The following core metrics were developed:

- **Total Revenue**
  → Sum of sales value across all transactions
- **Total Orders**
  → Count of unique order numbers
- **Total Quantity**
  → Total units sold
- **Total Customers**
  → Distinct count of customers
- **Average Order Value (AOV)**
  → Revenue per order

Total Revenue = SUMX (Sales, Sales [Quantity] * RELATED (Products [Unit Price USD]))

Total Orders = DISTINCTCOUNT (Sales [Order Number])

Total Quantity = SUM (Sales [Quantity])

Total Customers = DISTINCTCOUNT (Sales [Customer Key])

Average Order Value = DIVIDE ([Total Revenue], [Total Orders], 0)

KPI Design Logic

- KPIs were selected to reflect:
  - Business growth (Revenue, Orders)
  - Operational efficiency (AOV)
  - Customer scale (Customers)
- These KPIs are displayed as **cards in the dashboard** for quick executive-level insights.

Data Validation

- Measures were validated against raw data totals
- Cross-checks performed using:
  - Table visuals
  - Aggregation comparisons

👉 Ensures accuracy and reliability of insights

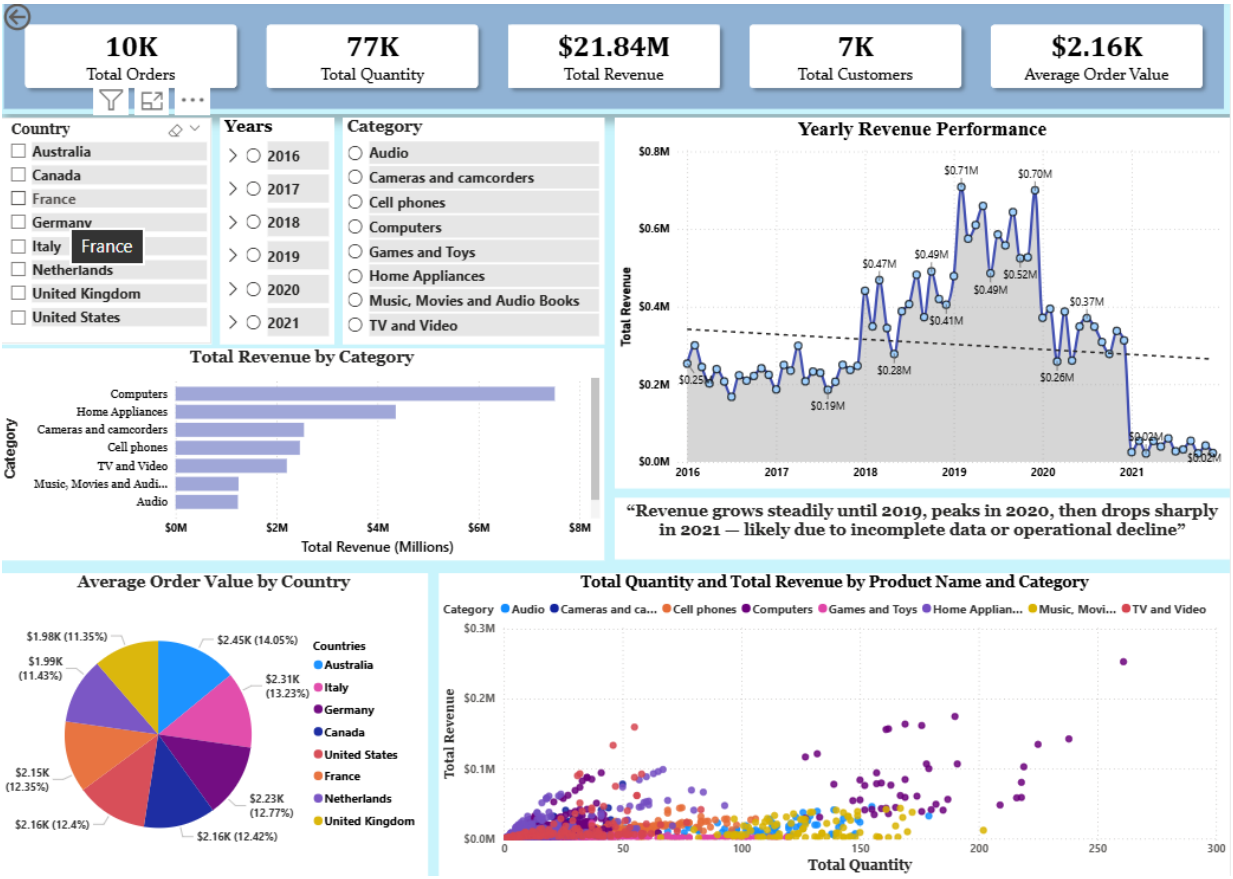
7. Dashboard Overview

Two dashboards were created:

Sales Performance Dashboard (Page 1)

Visuals Used

- KPI Cards:
  - Total Revenue ($21.84M)
  - Total Orders (10K)
  - Total Quantity (77K)
  - Customers (7K)
  - Average Order Value ($2.16K)
- Line Chart → Revenue Trend (Yearly)
- Bar Chart → Revenue by Category
- Scatter Plot → Quantity vs Revenue
- Country Analysis → Order Value Distribution



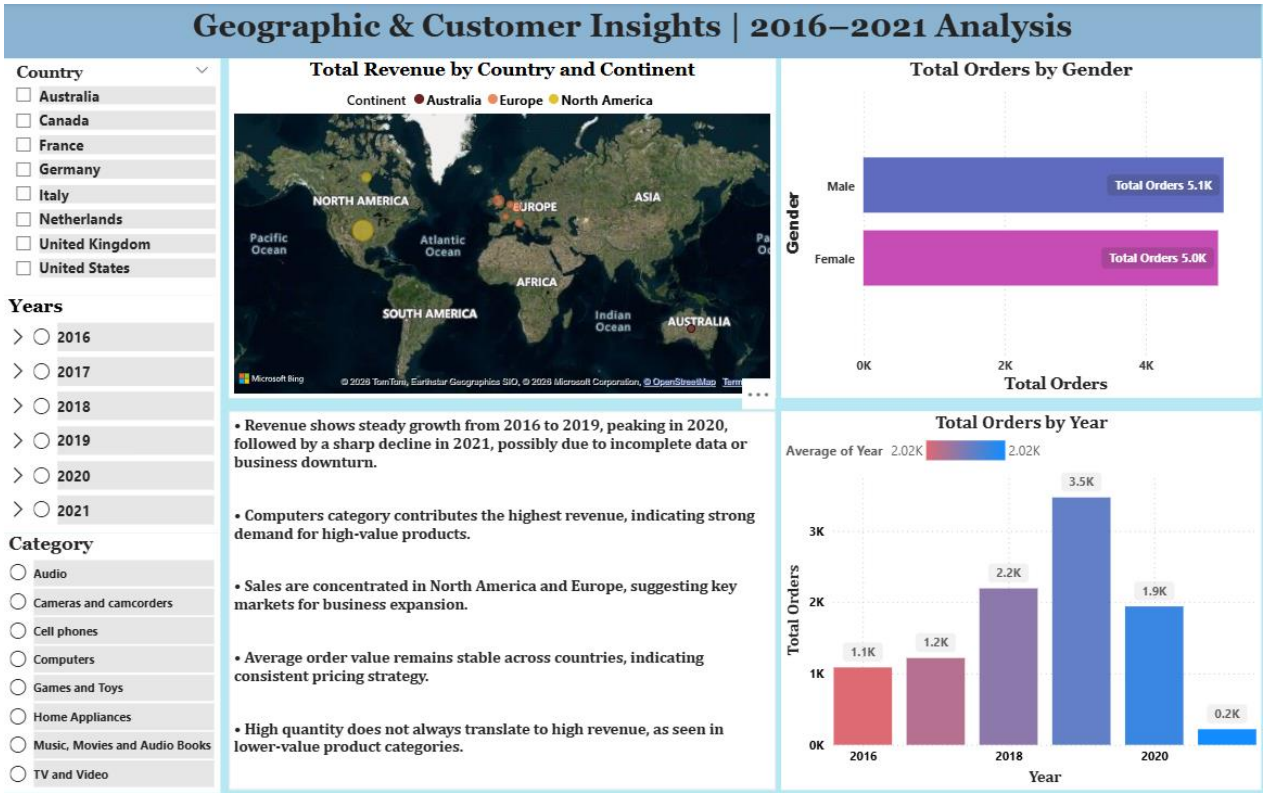
Geographic & Customer Insights Dashboard (Page 2)

Visuals Used

- Map → Revenue by Country & Continent
- Bar Chart → Orders by Gender
- Column Chart → Orders by Year
- Slicers:
  - Country
  - Year
  - Category

Interactive Features

- Drill-down capability
- Dynamic filtering
- Cross-filtering across visuals



8. Key Insights & Analysis

Revenue Trends

- Revenue increased steadily from **2016 to 2020**
- Peak performance observed in **2019–2020**
- Sharp decline in **2021**, indicating:
    - Possible data incompleteness
    - Or operational/business decline

Product Performance

- **Computers category** generates highest revenue
- High-value products contribute more to revenue
- Low-value categories show:
    - High quantity
    - Lower revenue impact

👉 Insight: Quantity ≠ Profitability

Customer Behaviour

- Orders are **almost evenly split by gender**
- Slight dominance of **male customers (~5.1K vs 5.0K)**

👉 Insight: No strong gender bias → broad market appeal

Geographic Insights

- Sales concentrated in:
    - **North America**
    - **Europe**

👉 Insight: These are primary revenue-driving regions

Sales Efficiency

- Scatter plot shows:
    - Some products have **high quantity but low revenue**
    - Others have **low quantity but high revenue**

👉 Insight: Focus on high-margin products, not just volume

Order Trends

- Orders peaked around **2019 (~3.5K orders)**
- Decline observed post-2020

👉 Insight: Possible market saturation or external disruption

9. Business Recommendations

Strategic Actions

- Focus on **high-revenue categories (Computers)**
- Expand operations in:
    - North America
    - Europe

Operational Improvements

- Investigate 2021 decline:
    - Data issue OR business issue

Product Strategy

- Promote high-margin products
- Optimize low-performing categories

Customer Strategy

- Maintain balanced targeting (gender-neutral strategy)

- Introduce loyalty programs for repeat customers

10. Conclusion

This project demonstrates:

- Strong **data modelling using star schema**
- Effective **Power BI dashboard design**
- Ability to extract **business-level insights**

The dashboards provide:

- Clear visibility into revenue trends
- Customer and geographic patterns
- Product performance

👉 Overall outcome: Data-driven decision-making for business growth